Mapping the Minds of LLMs: A Graph-Based Analysis of Reasoning LLM

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Abstract

Recent advances in test-time scaling have enabled Large Language Models (LLMs) to display sophisticated reasoning abilities via extended Chain-of-Thought (CoT) generation. Despite their potential, these Reasoning LLMs (RLMs) often demonstrate counterintuitive and unstable behaviors, such as performance degradation under few-shot prompting, that challenge our current understanding of RLMs. In this work, we introduce a unified graph-based analytical framework for better modeling the reasoning processes of RLMs. Our method first clusters long, verbose CoT outputs into semantically coherent reasoning steps, then constructs directed reasoning graphs to capture contextual and logical dependencies among these steps. Through comprehensive analysis across models and prompting regimes, we reveal that structural properties, such as exploration density, branching, and convergence ratios, strongly correlate with reasoning accuracy. Our findings demonstrate how prompting strategies substantially reshape the internal reasoning structure of RLMs, directly affecting task outcomes. The proposed framework not only enables quantitative evaluation of reasoning quality beyond conventional metrics but also provides practical insights for prompt engineering and the cognitive analysis of LLMs. Code and resources will be released to facilitate future research in this direction.

1 Introduction

Recent LLMs equipped with test-time scaling capabilities, such as OpenAI's o-series (OpenAI et al., 2024; OpenAI, 2025), DeepSeek-R1 (DeepSeek-AI et al., 2025), and Gemini-2.5 (Kavukcuoglu, 2025), employ a system II, *think-slow-before-answer*, pipeline that transforms how these models approach complex problems during test time. Rather than producing outputs directly after the input with normally limited token length, these reasoning models engage in explicit and free extended reasoning through Chain-of-Thought (Wei et al., 2022) mechanisms. This innovation enables reasoning models to decompose intricate challenges in various domains, explore multiple possible solutions, and self-assess intermediate conclusions before synthesizing final responses during their extended inference time. In general, these reasoning models currently outperform conventional LLMs on various types of benchmarks, which require advanced math (Patel et al., 2024) and coding (Jimenez et al., 2024) capability.

Despite these promising advancements, reasoning models exhibit undesire (Chen et al., 2024) and unstable (Yang et al., 2025b) behaviors that challenge the established understanding of large language models. One of the particularly striking phenomena is the performance degradation associated with few-shot learning, which in most cases improves the performance of conventional LLMs. Recent technique reports also documented that these RLMs are somehow more sensitive to prompts (DeepSeek-AI et al., 2025). We believe these existing unclear behaviors of RLM call for deeper investigations into how RLMs operate and reason.

Our research proposes a novel framework to trace the reasoning processes from a graph perspective. While some work has previously examined the correlation between the quantity of reasoning tokens and RLM's accuracy (Sui et al., 2025; Ballon et al., 2025; Yang et al., 2025b), our approach goes beyond the token perspective and focuses on the semantic organization of the model's reasoning processes. Specifically, our analytical frameworks first cluster raw and verbose reasoning tokens into coherent logic steps and then map their inter-dependencies as a graph, enabling globally semantical insights into how reasoning models reason at a higher level (Figure 1). After a comprehensive analysis of derived reasoning graphs, we identify specific quantifiable features that are associated

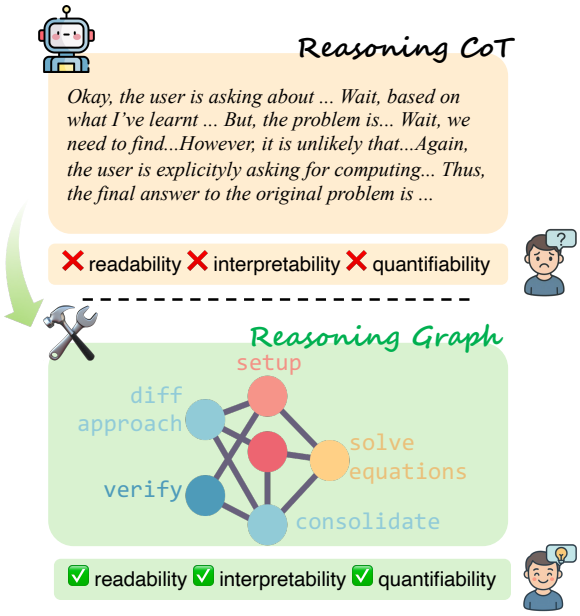

Figure 1: A conceptual overview of our framework for modeling the long reasoning CoT with a graph structure. This graph-based representation enables stronger readability for human researchers, systematic interpretability of the global structure, and quantifiable graph metrics for in-depth analysis.

with advanced reasoning behavior, which is often linked to higher problem-solving performance.

To summarize, our contributions in this paper include:

- a novel reasoning-graph toolkit that converts natural language long Chain-of-Thought into analyzable graph structures, enabling quantification of reasoning through topological and semantic metrics.
- comprehensive analysis of how different prompting strategies may influence reasoning LLMs, establishing quantitative boundaries for prompt engineering optimization.
- quantifiable indicators of reasoning quality beyond task accuracy, providing a higher-level cognitive understanding of reasoning in LLMs.

2 Related Works

Test-Time Scaling Similar to human dual-processing hypothesis of the mind (Da Silva, 2023), augmenting the computational budget at test-time has been shown to substantially enhance the reasoning capabilities of large language models (LLMs) (OpenAI et al., 2024; OpenAI, 2025; Kavukcuoglu, 2025; Anthropic, 2025). These reasoning LLMs (RLMs) show highly advanced self-reflection, backtracking, and cross-validation behavior during the extended chain-of-thought (CoT) responses, enabling them to tackle intricate reasoning challenges and outperform previous conventional base LLMs (Li et al., 2025; Chen et al., 2025).

Few-Shot Learning Few-shot prompting once emerged as a crucial technique for enhancing the performance and adaptability of large language models (LLMs) by providing limited yet highly informative demonstrations (Song et al., 2023). In detail, it leverages a minimal number of illustrative examples embedded directly into the input context, enabling models to rapidly generalize across diverse tasks without explicit parameter updates (Brown et al., 2020). However, many researchers and practitioners have reported that few-shot prompting could instead degrade the model's performance (DeepSeek-AI et al., 2025), signaling the instability of current reasoning LLMs. In this paper, we will examine the impact of zero/few-shot prompting on RLM's reasoning, assessing both the quality of internal reasoning and overall performance in in-context learning scenarios. Provide more valuable insights for future prompt engineering and model optimization.

Long CoT Analysis Some previous studies have identified a negative relationship between an RLM's accuracy and the number of reasoning tokens it generates (Ballon et al., 2025; Yang et al., 2025b). However, their analyses of RLMs mainly relied on a one-dimensional metric: the length of CoT token sequences. It still remains unclear and counterintuitive why even longer system II thinking could lead to performance degradation, suggesting a gap in our understanding of how RLMs work in general. In this work, we introduce a comprehensive structured framework to formulate the chain-of-thought process, offering deeper insights into the underlying reasoning behavior.

3 Constructing Reasoning Graph from Raw Reasoning Tokens

Given more computational resources at inference time, Reasoning Language Models (RLMs) can autonomously explore feasible solutions, perform cross-validations, actively access intermediate steps, and synthesize consolidations through

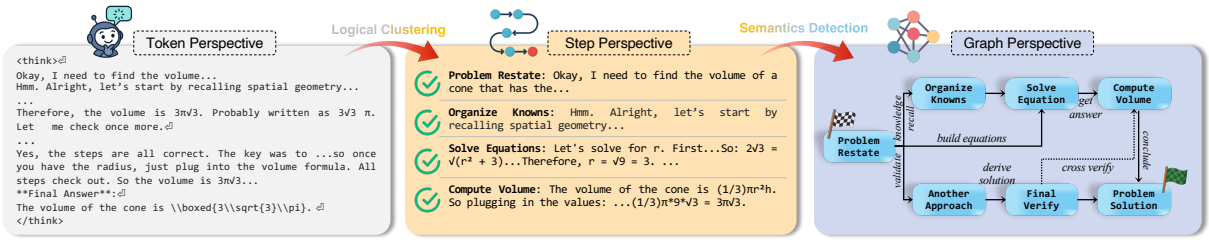

Figure 2: Our pipeline for building the graph structure from reasoning large language models' output. Starting from raw **token perspective**, we first use "\n\n" as natural delimiters to split the raw reasoning tokens into an ordered list of reasoning units. Then we perform logical clustering to combine logically cohesive reasoning units into a reasoning step (*node*), shifting into intermediate **step perspective**. Lastly, we detect semantical relationships (*edge*) between steps (*node*) to reveal the high-level **graph perspective** from reasoning LLM's output.

extended chain-of-thought tokens. This critical feature allows RLMs to fully release their internal reasoning potential under sophisticated challenges. However, current test-time scaling is also a double-edged sword: as models are encouraged to elaborate and reflect, their behavior often becomes unreliable and less predictable.

Counterintuitively, the "thinking out loud" style of RLM should, in theory, offer richer data for LLM interpretability research and help us understand how LLM actually reasons. Yet, to the best of our knowledge, there is still a lack of effective methods to systematically analyze and model the semantic content of RLM-generated reasoning tokens. To fill this gap, we propose a novel, structured approach for representing and dissecting the reasoning process of RLMs.

It is widely acknowledged that RLMs tend to generate complex, branching chains of thought. This pattern closely mirrors the way humans reason: rather than following a strictly linear path, our thinking often leaps between ideas, drawing on contextual cues, connecting prior knowledge and memories, searching for potential solutions, and constantly checking for errors along the way. It is precisely this interplay of multiple analytical paths that allows us to synthesize a coherent conclusion. Inspired by this convoluted human *mind map*, we propose a unified, graph-based framework (Figure 2) to model the structure of RLM outputs.

3.1 Graph Formalization

We can formally define the reasoning graph $G = (V, A)$ with the following components:

- $V = \{s_1, s_2, ..., s_n\}$: An ordered list of vertices representing semantically clustered reasoning steps.
- $A \in \{-1, 0, 1\}^{n \times n}$: An adjacency matrix representing the ternary logical relationship between reasoning steps.

In the remaining part of this section, we will first introduce a method for clustering long and verbose reasoning traces into discrete, semantically coherent reasoning steps, each of which will serve as a **node** in a reasoning graph (Section 3.2). We then describe how to extract semantic dependencies between these steps to form the **edges** of the reasoning graph ((Section 3.3)). This reasoning graph construction method will be used in subsequent sections as a key tool for quantitatively analyzing RLM's behavior.

3.2 Clustering Raw Tokens into Discrete Reasoning Steps

Long chain-of-thought (CoT) sequences generated by RLMs often span thousands of tokens. While these detailed traces offer rich insight into the model's reasoning process, their length and fragmented nature present challenges for systematic analysis. A common pre-processing strategy is to segment the output based on explicit delimiters: RLMs frequently insert the token "\n\n" to denote boundaries between successive thoughts. Let $T = (t_1, t_2, \ldots, t_N)$ represent the generated token sequence and use $\mathcal{D}$ = "\n\n" as the delimiter. We thus obtain an initial partition into *reasoning units*:

$$U = (u_1, u_2, \ldots, u_M),$$

where each u_i is a contiguous subsequence bounded by delimiters, i.e., $u_i = (t_{s_i}, \ldots, t_{e_i})$ and $t_{e_i+1} = \mathcal{D}$.

Despite its simplicity, delimiter-based segmentation has two fundamental limitations. For complex tasks such as advanced mathematical reasoning or